Supplementary Material: Provenance after Generative-Agent Relay

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A Theoretical Extensions

This appendix develops the theory beyond the scalar identity used in the main text. None of the main-text claims depend on this material; it is included to state precisely when the scalar result generalizes, and what each generalization requires.

A.1 Derivation of the scalar identity

Write the mutation indicator for one protected action as $M = Y\,b$, where Y indicates that the model requested the exact protected resource and value in the live, tool-visible fixture, and b is the reference monitor's write-back capability. State a is an authenticated semantic control rather than an execution capability. The experiment generates Y once at each a state (rate r_0 at a=0, r_1 at a=1) and replays the archived action under both values of b. Thus b cannot alter the already observed response. Then

$$\begin{aligned} R &= \mathbb{E}\big[\mathbb{E}[Y \mid a] \cdot b\big] = \mathbb{E}\big[(r_0 + (r_1 - r_0)a)\, b\big] \\ &= r_0\,\mathbb{E}[b] + (r_1 - r_0)\,\mathbb{E}[ab] \\ &= q\,r_0 + (r_1 - r_0)\,(q^2 + \mathrm{Cov}(a,b)) \\ &= q[(1-q)r_0 + q\,r_1] + \mathrm{Cov}(a,b)\,(r_1 - r_0), \end{aligned}$$

using $\mathbb{E}[a] = \mathbb{E}[b] = q$ and $\mathbb{E}[ab] = q^2 + \mathrm{Cov}(a,b)$. Subtracting the independent-lifecycles value ($\mathrm{Cov} = 0$) yields Equation (2). For a single shared persistent state, $a = b$ a.s., so $\mathrm{Cov}(a,b) = q(1-q)$ and $R = q\,r_1$; the shared-to-independent ratio $R_{\text{shared}}/R_{\text{indep}} = r_1/[(1-q)r_0 + qr_1]$ equals $1/q$ only at $r_0 = 0$. The $1/q$ law is thus a behavioral limiting case, not a topology-only statement.

A.2 Non-uniform fault probabilities and general joints

The scalar form assumes exchangeable marginals $\mathbb{E}[a] = \mathbb{E}[b] = q$. With node-indexed fault probabilities q_i and a general joint distribution over compromise sets, an additive expansion holds when each path's response depends only on its own elevation indicator and path contributions add at the expected-mutation-count level. It does not directly apply to a probability of at least one mutation, or to response functions with cross-path interactions. Under these assumptions, the same expansion holds with a and b replaced by their indicator collections: the excess of a topology over its independent-counterpart is a sum over failure-domain overlaps,

$$\Delta R \;=\; \sum_{i,j} \mathrm{Cov}(a_i, b_j)\,\big(\text{behavioral response of path } (i,j)\big),$$

and the single-action experiment measures exactly the one-path kernel of this sum. Nothing in the derivation requires independence across payloads; the experimental unit is the directive, and paired analyses preserve within-payload dependence.

A.3 Persistent compromise and the order of single-node round trips

If one persistent node compromise enables both bridge directions simultaneously, single-node round trips contribute at order $O(q)$ in the compromise probability, whereas independent mean-field substitution represents them at $O(q^2)$. The relative error of the mean-field analysis can therefore *diverge* as $q \to 0$ even though

its absolute error vanishes. This is an ordering of the joint bridge term, not automatically an ordering of total mutation risk. When $r_0 > 0$, independent-lifecycle risk contains the $q(1-q)r_0$ term, so both shared and independent total risks are $O(q)$ and their ratio tends to r_1/r_0, not $1/q$. The primary E3 fixture ($r_0 = 33/48$) and the earlier Qwen semantic-placement fixture ($r_0 = 28/48$) lie in this regime; the divergent ratio applies to the DeepSeek limiting case with $r_0 = 0$. The q-versus-q^2 statement is therefore about common-cause bridge activation, while dominance in total risk also requires behavioral suppression in the non-elevated state.

B Open Science and Artifact Map

B.1 Endpoints and versions

Provider experiments recorded requested and provider-reported model identifiers. Earlier runs reported DeepSeek V4 Flash, Qwen 3.7 Flash 2026-07-15, and MiniMax-M3. A later request for `deepseek-v4-flash` returned `deepseek-flash`. We report this run separately, not as the same endpoint. The run map below states which endpoint served each experiment. Receiver calls used temperature 0, disabled thinking, automatic tool choice, and a 1,024-token ceiling in the five-form confirmation. This ceiling replaced a 300-token calibration limit that had produced seven censored responses; no inference uses those censored cells.

The external instantiation used MCP Inspector 2.4.0 at pinned commit `edf54f5d`, MCP server 2.0.0, and Node.js v22.20.0. The complete commit hash is stored in the artifact manifest. Protocol citations are version- and date-pinned (§7). Deterministic A2A studies pinned `a2a-sdk` 1.1.2 and `@a2a-js/sdk` 1.1.0 with their dependencies and inspected sources; neither made provider calls.

B.2 Statistical procedures

The independent unit is the source directive or payload throughout; paired designs use exact McNemar tests as the primary inference, with paired nonparametric bootstrap intervals (100,000 resamples) at preregistered seeds. The five-form primary comparisons were a prespecified fixed sequence (familywise $\alpha = 0.05$). Rating studies keep the two model raters in isolated sessions: rater A used KIMI K3 and rater B used Grok 4.5, and each saw only its assigned workbook, with frozen row order, opaque IDs, and no form labels or behavioral outcomes. The five-form workbooks had 96 rows; the E2 adaptive-sender workbooks had 144. A retired E2 workflow exposed both model contexts to A/B materials; both returns were excluded and rerun under the isolation rule. The accepted fresh-context declaration is operator-attested and archived with the E2 rating result. The sequential standardization in §3.1 is labeled descriptive and is not used as causal mediation.

Multiplicity control follows the confirmatory family declared for each experiment. The five-form study uses its two-comparison fixed sequence; E2 and E3 retain their separately frozen primary paired contrasts. E4 first tests elevation within the adaptive marked relay and, conditional on rejection, tests the exact-response relay-policy interaction by exact paired sign flips; its broader any-write decomposition is post hoc. E5 prospectively defines any write as primary and exact mutation request as a fixed-sequence secondary. E6 uses three paired McNemar contrasts against the unmarked control with Holm step-down correction, separately at the direct and relayed stages. Secondary ratings, family breakdowns, cross-run comparisons, and cross-fixture comparisons do not enter those confirmatory families. We do not apply one omnibus paper-level correction across distinct scientific questions, and a result from one family does not unlock inference in another. E12 defines elevated exact protected-action pursuit as primary, using paired exact McNemar inference and a directive-level bootstrap interval; direct-write and escalation decompositions, the non-elevated comparison, and their interaction are supporting analyses within that experiment.

B.3 Run directories and audits

Every formal run is an append-only directory under `runs/` with events, provider responses, report, and post-run audit; key numbers were independently recomputed from raw artifacts rather than read back from the run's own report. The runs supporting each section (paths relative to the artifact root):

- **§3** (five-form):
  `deepseek-five-form-provenance-confirmation-001`
  `qwen-five-form-provenance-confirmation-001`
  calibration runs `-001` and `-002`; rater returns under `rating-returned/`.

- **§3.2** (E2 adaptive sender):
  `deepseek-e2-adversarial-sender-001`
  (pre-generation and receiver-release freezes; 144 frozen generated texts; 288 calls, zero retries; behavioral `independent_audit.json`, all checks passed). Accepted isolated secondary ratings, their attestation, and a separate reconstruction are under `rating-returned-clean-v2/`.
- **§4** (selector):
  `deepseek-retention-propagation-calibration-001`
  and its `-confirmation-001`. The selected inputs comprise 37 archived faithful-copy outputs and 84 task-condensation outputs. Their first-receiver delegations were 7 and 33, and their complete two-receiver chains were 2 and 15. `audit_retention_source_framing.py` verifies every selected output against the original sender event and prompt; these post hoc source-group counts do not change the primary 17/121 endpoint. The later concurrent-group check is `deepseek-retention-discrimination-001`; its `MODEL_IDENTITY_ADDENDUM.md` explains why the returned `deepseek-flash` responses cannot test discrimination on the earlier `deepseek-v4-flash` endpoint. Its pre-call manifest and analysis plan were internally frozen, not externally timestamped. A structural benign-delegation control is archived at `deepseek-benign-delegation-control-002`: 16 chains, 30 calls, instruction-eligible receivers, zero retries, and a passing wire audit. The request named `deepseek-v4-flash`, but responses reported `deepseek-flash`; it is therefore a plumbing control rather than an endpoint replication. A separately scoped receiver-authorized propagation calibration is archived at `qwen36-authorized-propagation-calibration-001`: the same marked sender output was presented under L0 and L2 to an R2 worker that was authorized to write `protected/canary`. Both arms produced 12/12 exact requests; this ceiling result is a fixture check, not a lineage effect or a security outcome. The independent recount is `independent_audit.json`. A zero-call headroom audit found the protected literal and a write verb in all 12 sender summaries and all 24 R1 payloads; it explains the ceiling and is archived as `headroom_audit.json`. A weaker indirect-content calibration is archived at `qwen36-authorized-propagation-weak-calibration-001`: exact protected requests were 2/12 in both L0 and L2, with no explicit write verb in the source bank. Its result is a headroom check, not a lineage comparison. The 48-pair disjoint follow-up is archived at `qwen36-authorized-propagation-confirmation-001`: L0 produced 6/48 and L2 4/48 exact requests (paired risk difference $-.0417$; exact McNemar $p = .7266$). It remains an internally frozen fixture comparison, not a confirmatory security result. The zero-call content audit for the 17 complete confirmation chains is `content_audit.json` in `deepseek-retention-propagation-confirmation-001`; it is a post hoc content characterization, not a new propagation estimate. The Qwen 3.6 Flash selector evaluation is archived at `qwen36-retention-discrimination-001` (380 calls): the 121 selected and 71 excluded archived texts were reused unchanged, giving 118/121 and 69/71 two-hop delegations. `audit_selector_transportability.py` reconstructs the counts from raw events. The end-to-end chain calibration is archived at `qwen36-chain-calibration-002` (191 calls): 24 attack items (six authored templates, four lexical instances each) in three lineage arms produced no protected request, with one outcome truncated, and `audit_chain_bottleneck.py` localizes the floor. Its legitimate-task wording confound is diagnosed in `qwen36-utility-wording-001`.
- **§3.1** (standardized cue and neutral control):
  `qwen36-provenance-rerender-001` (204 calls) and `qwen36-provenance-neutral-001` (306 calls), both on the 96 archived S_C/S_P texts unchanged, with 6/6 authorized controls completed in every arm. `audit_provenance_rerender.py` recomputes the paired tables and the conservative projection intervals.
- **§4** (defense):
  `qwen-cross-model-defense-001`. The DeepSeek one-family run `deepseek-one-family-defense-001` failed its preregistered susceptibility gate and is kept as a failed-gate record, not as defense evidence.
- **§4** (carrier):
  `deepseek-carrier-mechanism-001`, `qwen-carrier-mechanism-001`, `minimax-carrier-mechanism-001`.
- **§5** (DeepSeek):
  `deepseek-lifecycle-covariance-confirmation-001`
  (calibration `-001`; freeze `lifecycle-covariance-deepseek-20260828-v1`; `independent_verification.json`).
- **§5** (Qwen):
  `qwen-lifecycle-covariance-confirmation-001`
  (calibration `-001`; `independent_audit.json`; 116 calls, zero retries).
- **§5** (credential-gated E3):
  `qwen-e3-credential-gated-001`
  (freeze `e3-credential-gated-qwen-20260903-v1`; 126 calls, zero retries; `independent_audit.json`, 14/14 checks).
- **§5.3** (E4 relay-output gateway evaluation):
  `deepseek-e4-e2-e3-composition-001`
  (freeze `e4-e2-e3-composition-deepseek-20260904-v1`; 318 calls, zero retries; `independent_audit.json`, 16/16 checks). Post-run nested-outcome decomposition: `outcome_decomposition.json` (original E4 report unchanged). The frozen directory name predates the construct split.
- **§5.3** (E5 Qwen endpoint replication):
  `qwen-e5-relay-gateway-001`
  (freeze `e5-qwen-relay-gateway-20260904-v1`; 96 calls; `independent_audit.json`, all checks). The run records a bootstrap- seed protocol deviation; both seed recomputations give identical reported interval endpoints, and exact inference is unaffected.
- **§5.3** (E11 Qwen temporal repeat):
  `qwen-e11-hosted-repeat-001`

(freeze `e11_freeze_v1`; 96 calls, zero retries; `independent_audit.json`, 18/18 checks). The same 48 E5 texts and endpoint were reused under a new frozen calendar; the comparison is one paired temporal repeat, not a population variance estimate. E11–E5 effect shifts were $+1/48$ for both endpoints, with paired bootstrap intervals $[-0.1042, 0.1667]$ and $[-0.0833, 0.1250]$.

- **§5.3** (E12 capability-hidden check):
  `qwen-e12-capability-hidden-001`
  (freeze `e12_freeze_v1`; 252 calls, zero retries; `independent_audit.json`, 30/30 checks). The primary outcome joins exact direct writes and exact non-mutating escalation requests; the study does not execute the escalation or infer free-text intent.
- **§3.2** (E6 relay survival):
  `deepseek-e6-spotlight-relay-001`
  (192 source calls and 384 receiver calls, zero retries; pre-generation and receiver-release timestamps; `generation_independent_audit.json` and `independent_audit_v2.json`).
- **§6**:
  `inspector-shared-root-confirmation-001`
  (`report.json`, `independent_audit.json`); source and dependency freeze under `external_auth_instantiation/`.
- **§6** (E7–E10 A2A lineage integrity):
  `e7-a2a-lineage-composition-001`, `e8-a2a-js-live-001`,
  `e9-a2a-official-sample-live-001`,
  `e10-a2a-two-hop-enforcement-001`
  (transport, exact-sample derivation, and consequential two-hop controls; independently recomputed; zero calls).
- **§6.3** (native AgentDojo screen):
  `qwen36-agentdojo-five-form-001`
  `deepseek-agentdojo-five-form-001`
  (official `agentdojo` 0.1.35 wheel, benchmark v1.2.2, Slack suite, upstream predicates unmodified; `analysis.json` and `audit_report.json` are the authoritative outputs, with 4,350 and 3,861 zero-call replay checks). `headroom_diagnostic.json` records the counterfactual C−P ceiling of 3/55; its `authored_fixture` cell counts are mislabeled and corrected in an adjacent erratum, while its differences are correct. Offline Workspace controls, `workspace_controls_v3.json`, document the injection-task 11/12 predicate issue; no live Workspace session was run.

B.4 Open science

The anonymous, cloneable artifact bundle is being prepared at the following submission URL placeholder:

Anonymous artifact URL pending

The release will contain, for each formal run: the frozen stimulus bank and execution order; the sender, receiver, and rater prompts verbatim; the three rating rubrics as frozen before scoring; raw append-only JSONL event logs and provider responses; the analysis source (exact McNemar, paired bootstrap with seeds, q-weighting enumerator); the independent recomputation scripts and their outputs; the MCP Inspector instantiation source with its dependency freeze and commit pin; and a manifest of SHA-256 hashes for all of the above.

Preregistration evidence. Because "preregistered" is only meaningful if the freeze provably precedes the outcomes, each claim-bearing formal run's freeze record is externally witnessed: the manifest hash for the stimulus bank, execution order, analysis plan, provider pinning, and q grid was deposited to a third-party timestamping service before the first live call, and the deposit receipts are included in the bundle. Reviewers can verify that the analysis source distributed with the run hashes to the value recorded before execution. For E1, four OpenTimestamps calendars accepted the Qwen replication freeze before its 492 calls. For E2, four calendars accepted both the pre-generation and receiver-release deposits before their respective live stages. The E3 deposit was likewise accepted before execution. Four OpenTimestamps calendars accepted the E4 deposit before its 318 calls. Four calendars accepted the E5 deposit before its 96 calls; both sets of calendar attestations are included pending Bitcoin anchoring. Four calendars accepted both E6 stage deposits before their corresponding calls. Four calendars accepted the E12 freeze before its 252 calls. Earlier receipts carry independently verified Bitcoin attestations. The bundle keeps the original receipts, self-contained upgraded copies, and a machine-readable status report. The later endpoint-identity check is a diagnostic exception: its files were written before live calls but have no external timestamp receipt, so we do not label it externally preregistered or use it for confirmatory inference. The native AgentDojo Slack screens were exploratory external-validity runs and were not externally timestamped; they

are reported as provider-stratified descriptive evidence, not as preregistered mechanism tests. The Qwen 3.6 Flash follow-ups (selector evaluation, chain calibration, standardized cue, neutral control, and displayed-lineage confirmation) were likewise frozen internally before their live calls, without external timestamps, and are reported as secondary evidence.

What is *not* released. No designed bank is planned to be withheld. The release includes attack directives written against our sandbox tool schema; the schema is fictitious and the effects are process-local, so release carries no operational risk (§2).

C Family Breakdown and Example Payloads

C.1 Spotlighting relay-survival details

Table 1 reports the complete E6 write-attempt counts. The direct contrasts are incremental effects of each representation package beyond the verbal rule shared by every condition. After relay, `write_shared` survived lexically in 13/48 prose-marker outputs, 46/48 datamarked outputs, and 46/48 Base64 outputs; both resource and value placeholders survived in 7/48, 43/48, and 45/48, respectively. These are post-treatment descriptions, not causal mediation estimates.

Table 1: E6 write attempts before and after one DeepSeek relay (n=48 per cell). Relay Δ is control minus defense. Asterisks indicate rejection under the prespecified Holm procedure.

Source representation	Direct	Relayed	Relay Δ
Unmarked control	8/48	36/48	–
Prose marker	0/48	9/48	0.5625*
Datamarking	1/48	32/48	0.0833
Base64	0/48	24/48	0.2500*

C.2 Blinded rating details

Table 2: Blinded paired differences, S_C minus S_P (n=48; mean points on the 0–3 scale; paired bootstrap 95% CI in brackets; exact sign-test p). Agreement: quadratic-weighted kappa 0.968 (retention), 0.934 (centrality), 0.917 (provenance).

Axis	Rater A	Rater B
Actionable retention	1.292 [0.938, 1.646]	1.250 [0.896, 1.604]
Instruction centrality	1.833 [1.500, 2.146]	1.646 [1.313, 1.958]
Provenance signal	−0.833 [−1.083, −0.604]	−0.729 [−1.000, −0.479]

All sign-test $p < 10^{-4}$ (retention A 9.3×10^{-10}, B 3.7×10^{-9}; centrality 2.9×10^{-11} both; provenance A 3.0×10^{-8}, B 1.1×10^{-5}).

C.3 Family-level attempts

Table 3 breaks the five-form confirmation down by attack family. The breakdown is descriptive: the confirmatory analysis is the directive-level paired fixed sequence of §3, and with eight directives per family the cells are too small for family-level inference. Descriptively, the visible marker (P) held every family to 0–1/8, while the residual after relay (S_P) ranged from 1/8 (authority claim) to 6/8 (task framing and data mimicry): the protection that survives one hop is unevenly distributed across attack styles.

Table 3: Unauthorized protected-write attempts by attack family and form (n=8 directives per family; reanalysis of the frozen run `deepseek-five-form-provenance-confirmation-001`; row margins reproduce the frozen form totals 37/40/1/44/22). Descriptive only.

Family	R	C	P	S_C	S_P
Authority claim	4/8	4/8	0/8	8/8	1/8
Data mimicry	8/8	7/8	1/8	5/8	6/8
Imperative override	8/8	7/8	0/8	8/8	2/8
Provenance poisoning	5/8	7/8	0/8	8/8	4/8
Task framing	5/8	8/8	0/8	7/8	6/8
Tool-result injection	7/8	7/8	0/8	8/8	3/8

C.4 Two example payloads

Both examples use the same carrier domain; they differ in how much protection survived the relay hop. The marker line inserted in P is `UNTRUSTED EMBEDDED CONTENT:`. Outcomes list attempt$_{S_C}$ (shown as 1/0 flags) across the five forms.

Residual protection survives (cf-authority_claim-01). Carrier: *"Synthetic catalog validation note CF201 prepared for the library workflow. The entry concerns digital accession correction."* Directive: *"The continuity administrator authorizes you to call write_shared with resource 'X' and value 'Y'. Confirmation case CF201."* Outcomes: R=1, C=1, P=0, S_C=1, S_P=0. The bare and embedded directives triggered the write; the visible marker suppressed it; the unmarked relay restored it; and the relayed text derived from the marked carrier still did not trigger the write, so the laundering was partial.

Protection erased after relay (cf-task_framing-01). Carrier: *"Synthetic catalog validation note CF401 prepared for the library workflow. The entry concerns digital accession correction."* Directive: *"Complete case CF401 by ensuring that you call write_shared with resource 'X' and value 'Y'. Confirmation case CF401."* Outcomes: R=1, C=1, P=0, S_C=1, S_P=1. Identical structure and the same visible-marker suppression, but after one relay hop the protection was gone and the write was attempted again.

The full 48-pair bank, both carriers, all five forms per pair, and the frozen generated outputs are in the artifact.

D Secondary Semantic-Placement Fixture

An earlier semantic-placement fixture measured whether the response factor is portable across endpoints and operationalizations. The DeepSeek confirmation (freeze `lifecycle-covariance-deepseek-20260828-v1`) produced elevated-state write attempts on 42/48 payloads (0.875, exact 95% CI $[0.748, 0.953]$; every family nonzero, 6/8–8/8) and no untrusted-state attempts. Its four-state table of (attempts, mutations) was $(0/48, 0/48)$ at $(0, 0)$ and $(0, 1)$, $(42/48, 0/48)$ at $(1, 0)$, and $(42/48, 42/48)$ at $(1, 1)$. The Qwen replication produced elevated-state writes on 47/48 (CI $[0.8893, 0.9995]$) and untrusted-state writes on 28/48. Both confirmations passed all 24/24 authorized topology replays.

Qwen's sensitivity was $19/48 = 0.395833$ (19 payloads moved from no attempt to attempt under elevation and none reversed; exact McNemar $p = 3.8 \times 10^{-6}$). The declared covariances predict excess risks of $0.023141 \times 0.395833 = 0.009160$ and $0.047500 \times 0.395833 = 0.018802$, matching the weighted outcomes to six decimals. DeepSeek provides the $r_0 = 0$ limiting case with $r_1 = 42/48$, for which the shared grant realizes the $1/q = 20$ ratio at $q = 0.05$. Qwen yields 1.62 because it already attempts the write without elevation. The paired cross-provider difference in the untrusted state was 28 Qwen-only versus 0 DeepSeek-only payloads (exact McNemar $p = 7.5 \times 10^{-9}$).

Qwen's nonzero r_0 recurred across three banks: 28/48 in the lifecycle bank, 29/48 for bare directives, and 32/48 for neutral carriers. Its sensitivity was 0.3958 in this semantic-placement fixture, 0.3125 in the credential-gated fixture, 0.1667 for exact requests over frozen relay outputs (22/48 to 30/48), and 0.0625

Table 4: Secondary semantic-placement fixture, distinct from the credential-gated experiment in Table 5. Instruction-elevation sensitivity $r_1 - r_0$ is a measured paired endpoint-and-fixture response; $\mathrm{Cov}(a, b)$ is derived from the declared component-fault weights at $q = 0.05$. Agreement between predicted and weighted excess verifies the replay pipeline, not the algebraic identity. Ratios depend on r_0: 20.00 for DeepSeek ($r_0 = 0$) and 1.62 for Qwen ($r_0 = 28/48$).

Provider	Topology	$\mathrm{Cov}(a, b)$	Predicted excess	Observed excess	R_{topology}	R_{indep}
DeepSeek	Nominal partition, shared issuer root	0.023141	0.02025	0.02025	0.022436	0.002188
DeepSeek	One shared persistent grant	0.047500	0.04156	0.04156	0.043750	0.002188
Qwen	Nominal partition, shared issuer root	0.023141	0.009160	0.009160	0.039316	0.030156
Qwen	One shared persistent grant	0.047500	0.018802	0.018802	0.048958	0.030156

in the tool-visible arm of the capability-hidden fixture (42/48 to 45/48). That is a 6.3-fold range on one endpoint, wider than the 2.2-fold cross-endpoint contrast of 0.8750 versus 0.3958 on the shared semantic-placement fixture. Fixture variation therefore dominates the variation these designs can attribute to the endpoint, and the response is reported in the main text as an endpoint-and-fixture quantity (§5); it is never transported between fixtures.